

FINAL PROJECT TOPIC PROPOSAL

Smart Inventory and Packing Management System

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Java Console Application | Object-Oriented Programming

Brief Project Description

A console-based Java application for a small shop or warehouse. Multiple administrators and customers can authenticate; administrators maintain products and stock, while customers browse available inventory, place validated orders, and review order history. Users, products, and orders are restored from CSV files when the program runs again.

Core Features and Scope

INCLUDED

Role-based login for multiple ADMIN/CUSTOMER accounts; customer registration; product add, search, remove, stock display and update; order creation with quantity checks and automatic stock reduction; price, weight and packing-box calculation; order history; CSV persistence; input validation and exception handling.

NOT INCLUDED

Swing/JavaFX or other GUI, a database or cloud service, online payment, and other features outside the defined console application scope.

Technologies / Approach

Java console I/O with ten cooperating classes (Main, App, User, Product, InventoryManager, Order, OrderItem, OrderManager, FileManager and InputHelper) plus a Role enum. The design demonstrates encapsulation, composition, methods, control statements, ArrayList/HashMap collections, iteration, wrapper parsing, regular-expression validation, exceptions and CSV file input/output. Collections include users, products, orders and order items, with a HashMap product index.

Potential Implementation Challenges

Keeping product and order files consistent, preventing duplicate IDs and negative stock, reconnecting loaded order items to products, and recovering safely from invalid console input or malformed CSV rows.

Raw Time Estimation

Approximately 40 hours: design 5 hours, implementation 22 hours, testing/debugging 10 hours, and documentation/refinement 3 hours.

Evaluation Focus

The project solves a practical inventory problem and demonstrates the semester topics through clear class responsibilities, reusable methods, persistent data, multiple roles, cooperating objects and testable business rules.